

SIGNALS AFTER THE NOISE 2

OPERATOR ACTIVITY RECONSTRUCTION

WORKSPACE	INVESTIGATION WINDOW	ANCHOR	STATUS
LAW-Cyber-Range	2025-12-13 09:00–18:00 UTC	2025-12-13 09:48 UTC	CONFIRMED BREACH
PRIMARY HOST	SECONDARY HOST	COMPROMISED ACCOUNT	PHASES
azwks-phtg-01	azwks-phtg-02	vmadminusername	8 gated // 29 flags

EXECUTIVE SUMMARY

This hunt reconstructs operator activity on the PHTG estate following the confirmed initial access established in Hunt 05. The investigation window covers 13 December 2025 09:00–18:00 UTC, anchored at 09:48 UTC when the first post-intrusion session was recorded. The operator's activity consistently mimicked legitimate HealthCloud administration — nearly all of it looked like routine maintenance. The value of this hunt is in identifying deliberate intent hidden inside ordinary-looking telemetry.

The operator entered via credential reuse (T1078), moved laterally from azwks-phtg-02 to azwks-phtg-01, deployed a full persistence framework masquerading as HealthCloud services, established dual C2 channels to attacker-controlled infrastructure, tampered with Windows Defender, and concluded with a confirmed LSASS memory read — credential theft proven by telemetry.

29

FLAGS CAPTURED

8

HUNT PHASES

2

HOSTS COMPROMISED

280

REGISTRY MODIFICATIONS

22

HEALTHCHECK BEACONS

PHASE 01

Cold Trail

Credential reuse T1078 — not
brute force

PHASE 02

First Footsteps

_ps1 with Hidden+Bypass flags

PHASE 03

Quiet Roots

Run key + Startup LNK + Event log

PHASE 04

The Beacon Pair

PHTGHealthCloudSvc.exe +
encoded PS

PHASE 05

Outbound Whispers

updates/status.health-cloud.cc

PHASE 06

Doors Held Open

Defender exclusions
Cache+SvcExe

PHASE 07

Hands on the Vault

LSASS
ReadProcessMemoryApiCall

TIMESTAMP (UTC)	EVENT	DETAIL	MITRE
09:48:34	Lateral Movement	vmadminusername authenticates from 10.0.0.152 (azwks-phtg-02) to azwks-phtg-01	T1021.001
10:11:42	First Operator Script	_.ps1 executed: -WindowStyle Hidden -ExecutionPolicy Bypass. Temporary Defender exclusion added then removed	T1059.001
10:12:16	Tool Pull from C2	Outbound to updates.health-cloud.cc — ingress tool transfer	T1105
10:12:17	Payload Execution	PHTGHealthCloudSvc.exe /healthcheck /flag:FLAG-01 — beacon loop begins	T1071.001
10:12:18	Defender Exclusion	Add-MpPreference -ExclusionPath Cache -Force	T1562.001
10:12:27	Attribute Hiding	attrib +h +s on Cache (17 ops) and TempCache (3 ops)	T1564.001
10:13:00	Startup Persistence	PHTG HealthCloud.lnk dropped in Startup folder	T1547.001
10:13:01	Run Key Persistence	PHTGHealthCloudTray → HealthCloudTray.ps1 written to HKCU Run key	T1547.001
10:13:43	Encoded Beacon 1	Invoke-WebRequest to status.health-cloud.cc/api/checkin?flag=FLAG-09	T1027
10:13:56	Encoded Beacon 2	Invoke-WebRequest to status.health-cloud.cc/api/status?flag=FLAG-10	T1027
10:14:10	AMSI Probe	amsi_probe.ps1 executed — testing AV scan interface before main payload	T1562.001
10:14:29	Process Exclusion	Add-MpPreference -ExclusionProcess PHTGHealthCloudSvc.exe -Force	T1562.001
10:14:37	LSASS Handle — Query	OpenProcessApiCall lsass.exe DesiredAccess 0x1410 (PROCESS_QUERY_INFORMATION)	T1003.001
10:14:38	LSASS Handle — Full Access	OpenProcessApiCall lsass.exe DesiredAccess 0x1F0FFF (PROCESS_ALL_ACCESS)	T1003.001
10:14:38	LSASS Memory Read CONFIRMED	ReadProcessMemoryApiCall — credentials dumped from lsass.exe memory	T1003.001
10:15:15	Lineage Break	cmd.exe /c powershell.exe hc_lineage.ps1 — parent-child lineage broken	T1059.003
10:16:09	Batch Launcher	cmd.exe /c phtg_health_diag_update_FLAG-22.bat — final stage execution	T1059.003

FIELD	VALUE	SIGNIFICANCE
Source Host	azwks-phtg-02	Initial access host from Hunt 05
Source IP	10.0.0.152	Private IP of azwks-phtg-02
Target Host	azwks-phtg-01	Secondary host — operator's main workspace
Account	vmadminusername	Credential reuse — same creds across fleet
LogonType	Network then RemoteInteractive	Tool auth then full RDP session
Time	09:48:34–09:48:40 UTC	Anchor moment — 6 seconds to establish session
Onward Movement	None	Blast radius contained to two hosts
Vector	T1078 Valid Accounts / Credential Reuse	NOT brute force — credentials known in advance

MECHANISM	ARTEFACT	COMMAND / PATH	STATUS
Run Key	PHTGHealthCloudTray	powershell.exe -NoProfile -WindowStyle Hidden -ExecutionPolicy Bypass -File C:\ProgramData\PHTG\HealthCloud\Bin\HealthCloudTray.ps1	FIRE D x2
Startup Folder	PHTGHealthCloud.lnk	C:\Users\vmAdminUsername\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\	DETECTED NOT BLOCKED
Event Log Reg	PHTGHealthCloud	HKLM\SYSTEM\ControlSet001\Services\EventLog\Application\PHTGHealthCloud	ACTIVE
Defender Excl	Cache path + SvcExe	ExclusionPath: Cache // ExclusionProcess: PHTGHealthCloudSvc.exe	ACTIVE

OPERATOR SIGNATURE FLAGS — Present on Every Deliberate Execution

FLAG	VALUE	PURPOSE
-NoProfile	(no argument)	Skip PowerShell profile — faster, avoids profile-based logging
-WindowStyle	Hidden	No visible window — user never sees execution
-ExecutionPolicy	Bypass	Override execution restrictions — run unsigned scripts

DOMAIN	ROLE	FIRST SEEN	MECHANISM
updates.health-cloud.cc	Tool delivery / payload pull	10:12:16 UTC	Invoke-WebRequest (plain PS)
status.health-cloud.cc	Beacon callbacks / check-in	10:13:44 UTC	Invoke-WebRequest (-EncodedCommand)
raw.githubusercontent.com	Late-stage tool pull	16:24:46 UTC	LOL delivery via GitHub raw

DUAL CHANNEL BENEFIT

BENEFIT	EXPLANATION
Resilience	If one channel is blocked or detected, the other continues operating independently — the operator never goes dark
Detection Splitting	Two low-volume channels across different mechanisms split the traffic signature — neither alone crosses the detection threshold that triggers a response

CREDENTIAL THEFT — LSASS

The operator executed **task_FLAG-13.ps1** from the Defender-excluded Cache directory at 10:14:37 UTC. This script made two sequential calls to open an LSASS process handle, escalating access rights within one second, followed immediately by a confirmed memory read.

TIME	ACTIONTYPE	DESIRED ACCESS	HEX	MEANING
10:14:37	OpenProcessApiCall	5136	0x1410	PROCESS_QUERY_INFORMATION — probe, low privilege
10:14:38	OpenProcessApiCall	2047999	0x1F0FFF	PROCESS_ALL_ACCESS — full unrestricted access
10:14:38	ReadProcessMemoryApiCall	—	—	CONFIRMED: memory read — credentials dumped

HOST IOCs

TYPE	VALUE	DESCRIPTION	SEVERITY
PROCESS	PHTGHealthCloudSvc.exe	Masquerades as bitsadmin.exe — beacon loop	CRITICAL
SCRIPT	HealthCloudTray.ps1	Persistence script — Run key payload	HIGH
SCRIPT	amsi_probe.ps1	AMSI detection probe before main payload	HIGH
SCRIPT	hc_lineage.ps1	Lineage enumeration via cmd.exe wrapper	HIGH
SCRIPT	_.ps1	Master operator script — Hidden+Bypass execution	CRITICAL
FILE	PHTG HealthCloud.lnk	Startup folder persistence — detected not blocked	HIGH
PATH	C:\ProgramData\PHTG\HealthCloud\Cache	Hidden artefact directory — attrib +h +s	HIGH
REGKEY	HKCU\...\CurrentVersion\Run\PH TGHealthCloudTray	Run key persistence entry	CRITICAL
ACCOUNT	vmadminusername	Compromised account — used across both hosts	CRITICAL

NETWORK IOCs

TYPE	VALUE	ROLE	SEVERITY
FQDN	updates.health-cloud.cc	Tool delivery C2	CRITICAL
FQDN	status.health-cloud.cc	Beacon callback C2	CRITICAL
DOMAIN	health-cloud.cc	Parent attacker domain (.cc TLD)	CRITICAL
IPv4	10.0.0.152	azwks-phgt-02 internal — lateral movement source	HIGH


```
// LATERAL MOVEMENT DETECTION – Credential Reuse Pattern
```

```
DeviceLogonEvents
| where TimeGenerated between (datetime(2025-12-13T09:00:00) .. datetime(2025-12-13T18:00:00))
| where DeviceName == "azwks-phtg-01"
| where ActionType == "LogonSuccess"
| where LogonType in ("RemoteInteractive", "Network")
| summarize
FailedAccounts = make_set_if(AccountName, ActionType == "LogonFailed"),
SuccessAccounts = make_set_if(AccountName, ActionType == "LogonSuccess"),
FailCount = countif(ActionType == "LogonFailed"),
SuccessCount = countif(ActionType == "LogonSuccess")
by RemoteIP
| where SuccessCount > 0
| order by SuccessCount desc
```

```
// OPERATOR SIGNATURE FLAGS – PowerShell Concealment Hunt
```

```
DeviceProcessEvents
| where DeviceName == "azwks-phtg-01"
| where TimeGenerated > datetime(2025-12-13T09:48:40Z)
| where FileName =~ "powershell.exe"
| where ProcessCommandLine has_any (
"-WindowStyle Hidden",
"-ExecutionPolicy Bypass",
"-EncodedCommand",
"-NoProfile")
| project TimeGenerated, ProcessCommandLine,
InitiatingProcessFileName, InitiatingProcessAccountName
| order by TimeGenerated asc
```

```
// DECODE ENCODED POWERSHELL BEACONS – Inline KQL Decode
```

```
DeviceProcessEvents
| where DeviceName == "azwks-phtg-01"
| where FileName =~ "powershell.exe"
| where ProcessCommandLine contains "-EncodedCommand"
| extend EncodedBlob = extract(
@"-EncodedCommand\s+([A-Za-z0-9+/=]+)", 1, ProcessCommandLine)
| extend DecodedCommand = base64_decode_tostring(EncodedBlob)
| project TimeGenerated, ProcessCommandLine, DecodedCommand
| order by TimeGenerated asc
```

```
// RUN KEY PERSISTENCE DETECTION – Operator Payload in Run Key
```

```
DeviceRegistryEvents
| where RegistryKey contains "CurrentVersion\\Run"
| where ActionType == "RegistryValueSet"
| where RegistryValueData has_any (
```

```
"-WindowState Hidden",  
"-ExecutionPolicy Bypass",  
"-NoProfile")  
| project TimeGenerated, DeviceName,  
RegistryValueName, RegistryValueData  
| order by TimeGenerated asc
```

```
// LSASS CREDENTIAL DUMP DETECTION – Full 3-Step Chain
```

```
// Step 1: Find anomalous LSASS access (non-system account)
```

```
DeviceEvents
| where DeviceName == "azwks-phgt-01"
| where ActionType == "OpenProcessApiCall"
| where FileName =~ "lsass.exe"
| where InitiatingProcessAccountName !in (
"system", "network service", "local service")
| extend DesiredAccess = tostring(parse_json(AdditionalFields).DesiredAccess)
| project TimeGenerated, InitiatingProcessAccountName,
InitiatingProcessFileName, DesiredAccess
| order by TimeGenerated asc
```

```
// Step 2: Confirm memory read fired after handle access
```

```
DeviceEvents
| where DeviceName == "azwks-phgt-01"
| where FileName =~ "lsass.exe"
| where InitiatingProcessAccountName =~ "vmadminusername"
| summarize count() by ActionType
| order by count_ desc
```

```
// DEFENDER TAMPERING – Exclusions and Mode Changes
```

```
// Add-MpPreference exclusion detection
```

```
DeviceEvents
| where ActionType == "PowerShellCommand"
| extend Command = tostring(parse_json(AdditionalFields).Command)
| where Command has_any (
"Add-MpPreference", "Remove-MpPreference",
"ExclusionPath", "ExclusionProcess",
"DisableRealtimeMonitoring")
| project TimeGenerated, DeviceName, Command
| order by TimeGenerated asc
```

```
// FILENAME vs ORIGINAL FILENAME MISMATCH – LOLBin Masquerade
```

```
// Catch executables claiming to be something they are not
```

```
DeviceProcessEvents
| where DeviceName == "azwks-phgt-01"
| where FileName != ProcessVersionInfoOriginalFileName
| where isnotempty(ProcessVersionInfoOriginalFileName)
| where InitiatingProcessAccountName =~ "vmadminusername"
| project TimeGenerated, FileName,
ProcessVersionInfoOriginalFileName,
ProcessCommandLine, FolderPath
| order by TimeGenerated asc
```

PRIORITY	ACTION	RATIONALE
IMMEDIATE	Isolate both azwks-phtg-01 and azwks-phtg-02	Lateral movement confirmed — both hosts are compromised
IMMEDIATE	Revoke vmadminusername across entire fleet	Credential reuse confirmed — same creds worked on both hosts
IMMEDIATE	Remove Run key entry PHTGHealthCloudTray	Persistence confirmed fired twice — still active
IMMEDIATE	Delete PHTG HealthCloud.Ink from Startup folder	Second persistence mechanism — detected but not blocked
IMMEDIATE	Remove Defender exclusions for Cache and PHTGHealthCloudSvc.exe	Exclusions allow payload to run undetected
IMMEDIATE	Block health-cloud.cc at DNS and perimeter	Both C2 subdomains resolve under this parent domain
IMMEDIATE	Assume credentials from azwks-phtg-01 are compromised	ReadProcessMemoryApiCall on lsass.exe confirmed — all creds in memory are stolen
SHORT TERM	Enforce Defender active mode via GPO — no passive mode	Defender detected PHTG HealthCloud.Ink but took no action in passive mode
SHORT TERM	Alert on ProcessCommandLine with -WindowStyle Hidden + -ExecutionPolicy Bypass	Operator signature — every deliberate execution carried these flags
SHORT TERM	Alert on FileName != ProcessVersionInfoOriginalFileName under user accounts	PHTGHealthCloudSvc.exe claimed to be bitsadmin.exe — masquerade detection
LONG TERM	Implement LSASS Protected Process Light (PPL)	PPL prevents userland processes from opening LSASS handles entirely
LONG TERM	Deploy Credential Guard on all workstations	Credential Guard moves credentials to isolated VSM — ReadProcessMemory returns nothing
LONG TERM	Alert on OpenProcessApiCall to lsass.exe from non-system accounts	DesiredAccess 0x1F0FFF (PROCESS_ALL_ACCESS) from user context is always malicious